

SOVEREIGN QUANTUM-HFT CORE ARCHITECTURE

Comprehensive 44-Module Technical Specifications & Component Index

MODULE SUITE 1: LOW-LATENCY NETWORKING & LAYER 3 INGESTION

- IngestionDepthEngine.cpp: High-performance Level 3 book parser aligned to 64-byte boundaries.
- L3OrderNode.h: Thread-safe data structure nodes optimizing L1/L2 cache line propagation.
- LockFreeSPSCRingBuffer.h: Single-Producer Single-Consumer lock-free ring arrays for memory isolation.
- FIX44ProtocolParser.cpp: Bare-metal execution engine optimized for raw FIX protocol translation.
- SocketLayerGateway.cpp: Zero-copy asynchronous network interface controlling direct socket queues.

MODULE SUITE 2: QUANTUM ENTRAPY & MATHEMATICAL FILTERS

- InstitutionalQuantumEntanglementEngine.py: Singular Value Decomposition matrix parser computing live Von Neumann Entropy.
- UniversalQuantumKineticMatrix.py: Predictive matrix tracker evaluating structural return arrays.
- ShannonEntropyCalculators.cpp: Low-overhead real-time information metric processing at the OS layer.
- SupremeChaosFractalEngine.cpp: Real-time phase space reconstruction evaluating negative Lyapunov exponents.
- LocalizedFeatureMatrix_kNN.py: Native k-NN predictive engine pinned to isolated CPU cores.

MODULE SUITE 3: ASYNCHRONOUS RISK ENGINE & ANTI-TRAP SECURITY

- MasterAntiTrapEngine.cpp: Real-time pre-execution firewall isolating toxic broker spreads and spread spikes.
- MasterLogicalOrchestrator.cpp: Blended decision core fusing institutional order flow with technical variables.
- Quant_Risk_Executor.mq5: Hybrid production gateway enforcing programmatic 3.5% daily drawdown limits.
- UltimateStealthHftEngine.cpp: RNG-driven volume randomization engine obfuscating trade placement patterns.
- VolatilityClusteringShield.cpp: Dynamic execution break interchanging trading bounds during liquidity shocks.

MODULE SUITE 4: INTEGRATION INTERFACES & EXECUTION CHANNELS

- Quant_Pipe_Chart.mq5: Asynchronous inter-process memory bridge syncing Python models with MT5 terminals.
- Quant_UI_Dashboard.mq5: High-throughput zero-lag graphical interface rendering real-time core metrics.
- MultiAssetExecutionBridge.py: Parallel thread pipeline managing cross-exchange commodities and forex pairs.
- ProductionBuildGateway.h: Native compiled C++ DLL interface unifying system runtimes for live deployment.